

SQL Business Report: Chinook Music Store Analytics

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Tool Used: MySQL Workbench

1. Project Overview

Goal:

To derive meaningful business insights from a relational music store database (Chinook) using advanced SQL techniques such as JOINS, subqueries, and window functions.

Tools Used:

- MySQL Workbench for query writing and execution
- Chinook Dataset (in CSV format) as the data source

Dataset Summary:

The Chinook database contains information on:

- Tracks and their pricing, genre, and album associations
- Artists and albums
- Customers, invoices, and invoice lines
- Employee support representatives
- Geographic data including customer countries

2. SQL Environment Setup

Database Creation:

```
CREATE DATABASE chinook;  
USE chinook;
```

Data Import:

CSV files were imported using Table Data Import Wizard in MySQL Workbench:

- Album.csv
- Artist.csv
- Track.csv
- Invoice.csv
- Customer.csv
- InvoiceLine.csv
- Genre.csv
- Employee.csv
- Mediatype.csv

- Playlist.csv
- Playlisttrack.csv

Each file was saved in UTF-8 format to prevent encoding issues.

Table Creation & Mapping:

Each CSV was mapped to a table with appropriate column types (e.g., INT, VARCHAR, DECIMAL). Primary and foreign keys were respected for relational integrity.

3. KPI Analysis

- **Total Revenue:**
SELECT SUM(Total) AS TotalRevenue FROM Invoice;
Insight: Total revenue generated from all customer purchases.
- **Top 5 Customers by Revenue:**
SELECT c.CustomerId, c.FirstName, c.LastName, SUM(i.Total) AS Revenue
FROM Customer c
JOIN Invoice i ON c.CustomerId = i.CustomerId
GROUP BY c.CustomerId, c.FirstName, c.LastName
ORDER BY Revenue DESC
LIMIT 5;
Insight: Identifies high-value customers for loyalty programs.
- **Average Sales per Customer:**
SELECT AVG(Total) AS AverageInvoiceValue FROM Invoice;
Insight: Useful for setting benchmarks in sales strategy.

4. Business Insights from Joins, Subqueries & Windows

1. **JOIN Query – Track Names with Artist and Genre:**
SELECT t.Name AS TrackName, ar.Name AS Artist, g.Name AS Genre
FROM Track t
JOIN Album a ON t.AlbumId = a.AlbumId
JOIN Artist ar ON a.ArtistId = ar.ArtistId
JOIN Genre g ON t.GenreId = g.GenreId;
Insight: Useful for content curation and catalog display.

2. **Subquery – Customers Above Average Spend:**

```
SELECT FirstName, LastName, Total
FROM Customer c
JOIN Invoice i ON c.CustomerId = i.CustomerId
WHERE Total > (SELECT AVG(Total) FROM Invoice);
```

Insight: Helps segment premium customers for retention campaigns.

3. **Window Function – Rank Tracks by Price within Genre:**

```
SELECT t.Name, g.Name AS Genre,
RANK() OVER (PARTITION BY g.Name ORDER BY t.UnitPrice DESC) AS PriceRank
FROM Track t
JOIN Genre g ON t.GenreId = g.GenreId;
```

Insight: Shows top premium tracks per genre for targeted promotions.

Deep-Dive KPIs

1) **Top-Selling Track by Revenue:**

```
SELECT t.Name AS TrackName,
SUM(il.UnitPrice * il.Quantity) AS TotalRevenue
FROM InvoiceLine il
JOIN Track t ON il.TrackId = t.TrackId
GROUP BY t.TrackId, t.Name
ORDER BY TotalRevenue DESC
LIMIT 5;
```

2) **Artist with the Most Tracks:**

```
SELECT ar.Name AS ArtistName,
COUNT(t.TrackId) AS TrackCount
FROM Artist ar
JOIN Album al ON ar.ArtistId = al.ArtistId
JOIN Track t ON al.AlbumId = t.AlbumId
GROUP BY ar.ArtistId, ar.Name
ORDER BY TrackCount DESC
LIMIT 5;
```

3) **Customer Lifetime Value:**

```
SELECT c.CustomerId, c.FirstName, c.LastName, SUM(i.Total) AS LifetimeValue
FROM Customer c
JOIN Invoice i ON c.CustomerId = i.CustomerId
GROUP BY c.CustomerId, c.FirstName, c.LastName
ORDER BY LifetimeValue DESC;
```

4) **Revenue by Country:**

```
SELECT c.Country, SUM(i.Total) AS TotalRevenue
FROM Invoice i
JOIN Customer c ON i.CustomerId = c.CustomerId
GROUP BY c.Country
ORDER BY TotalRevenue DESC;
```

5) **Average Items per Invoice:**

```
SELECT AVG(ItemsPerInvoice) AS AvgItemsPerInvoice
FROM (
  SELECT DISTINCT il.InvoiceId,
  COUNT(il.TrackId) OVER (PARTITION BY il.InvoiceId) AS ItemsPerInvoice
  FROM InvoiceLine il
) AS Sub;
```

Conclusion

Using SQL, we extracted key business insights from the Chinook database. Advanced functions like JOINS, subqueries, and window functions enabled deep analysis across customer, artist, product, and geographic dimensions.